

Machine Learning–Based Estimation of Wireless Coverage Maps Using Terrain Data and GPS-Sampled Signal Measurements

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Abstract—Accurate radio coverage estimation is a key requirement for the deployment and optimization of Low Power Wide Area Networks (LPWANs) such as LoRaWAN. Conventional propagation models are attractive because of their simplicity, but their predictive performance often degrades in heterogeneous environments where terrain morphology, land use, and local propagation conditions significantly influence signal behavior.

This paper presents a data-driven wireless coverage prediction framework based on real-world LoRaWAN measurements and geographic information system (GIS) data. A dataset containing 10,203 RSSI measurements collected across two Vietnamese cities, Da Nang and Hai Phong, was used to train and evaluate the proposed model. The framework combines radio transmission parameters, geometric features, terrain elevation descriptors, obstruction and Fresnel-zone metrics, land-use information, and nearest-neighbor signal statistics extracted from a spatial reference database.

Several machine learning algorithms were investigated, and an optimized Extra Trees regressor was selected as the final prediction engine. Experimental results obtained using five-fold cross-validation demonstrate strong predictive performance, achieving an average coefficient of determination of $R^2 = 0.9285$, a mean absolute error of 1.93 dBm, and a root mean square error of 2.92 dBm. Feature importance analysis further highlights the contribution of terrain-aware and neighbor-based descriptors to coverage prediction accuracy.

To facilitate practical deployment, the trained model was integrated into a dedicated inference engine capable of generating RSSI predictions at arbitrary geographic locations without retraining. The proposed approach provides an accurate and scalable solution for wireless coverage mapping and network planning in LPWAN environments.

Index Terms—RSSI, Wireless Coverage Estimation, Machine Learning, LoRaWAN, LPWAN, Terrain-Aware Propagation

I. INTRODUCTION

Low Power Wide Area Networks (LPWANs) have emerged as a fundamental communication technology for Internet of Things (IoT) applications including smart cities, environmental monitoring, industrial automation, precision agriculture, and large-scale sensing systems. Among these technologies, LoRaWAN has gained significant attention due to its ability to provide long-range communication while maintaining very low energy consumption. Accurate coverage prediction is therefore essential for gateway deployment, network planning, capacity estimation, and quality-of-service optimization.

Predicting radio signal propagation remains a challenging problem because wireless channels are strongly influenced by environmental conditions. Signal attenuation depends not only on the transmitter–receiver distance but also on terrain elevation, vegetation, buildings, land use, and local obstructions. As a result, coverage estimation methods must account for a large number of interacting factors that vary across geographical regions.

Traditional propagation models such as Okumura–Hata, COST-231, ECC-33, and the Irregular Terrain Model (ITM) are widely used for wireless network design. These models provide computationally efficient estimates based on empirical or semi-empirical formulations. However, their assumptions often fail to accurately represent complex propagation environments, particularly in urban areas, mountainous regions, or mixed land-use scenarios. Consequently, prediction errors may become significant when these models are applied outside the environments for which they were originally calibrated.

Recent advances in machine learning have enabled the development of data-driven propagation models capable of learning nonlinear relationships directly from field measurements. Previous studies have demonstrated the effectiveness of algorithms such as Random Forests, Support Vector Machines, Artificial Neural Networks, Gradient Boosting, and Deep Neural Networks for RSSI and path-loss prediction. Compared with conventional propagation models, these approaches can better exploit environmental information and adapt to local propagation characteristics.

Despite these advances, several challenges remain. Many existing approaches rely on simulation-generated datasets rather than real-world measurements, require computationally intensive geographic processing, or focus exclusively on terrain-related features while neglecting local spatial signal correlations. Furthermore, practical deployment considerations are often overlooked, making it difficult to integrate trained models into real-time coverage mapping applications.

To address these limitations, this paper proposes a terrain-aware and neighbor-assisted machine learning framework for LoRaWAN RSSI prediction. The proposed approach combines radio transmission parameters, geometric descriptors, terrain elevation characteristics, obstruction and Fresnel-zone metrics, land-use information, and nearest-neighbor signal statistics derived from a spatial reference database. The framework is trained using 10,203 real-world LoRaWAN measurements collected across two Vietnamese cities, Da Nang and Hai Phong, covering a wide range of propagation conditions and

gateway configurations.

An optimized Extra Trees regression model is adopted as the final prediction engine. Beyond predictive accuracy, particular attention is given to deployment feasibility through the development of a dedicated inference architecture capable of generating RSSI predictions at arbitrary GPS coordinates using only a pre-trained model and a compact spatial reference database.

The main contributions of this work are fourfold. First, a real-world LoRaWAN propagation dataset comprising 10,203 RSSI measurements collected across two Vietnamese cities and 13 gateways was constructed. Second, a terrain-aware feature engineering framework integrating geometric, topographic, obstruction, Fresnel-zone, land-use, and neighbor-based spatial descriptors was developed. Third, an Extra Trees regression model was optimized for RSSI prediction, achieving an average R^2 of 0.9285, a MAE of 1.93 dBm, and a RMSE of 2.92 dBm. Finally, a dedicated inference engine was designed for practical integration into wireless coverage mapping and network planning applications.

The remainder of this paper is organized as follows. Section II reviews related work on radio propagation modeling and machine learning-based coverage prediction. Section III presents the dataset, feature engineering methodology, and proposed prediction framework. Section IV describes the experimental setup and evaluation methodology. Section V discusses the obtained results and compares the proposed approach with existing methods. Finally, Section VI concludes the paper.

II. RELATED WORK

Accurate wireless coverage prediction has long been a fundamental problem in radio network planning and optimization. Coverage estimation is typically performed either through analytical and empirical propagation models, interpolation techniques, or more recently, data-driven machine learning approaches. The increasing complexity of wireless environments, particularly in heterogeneous urban and suburban deployments, has motivated the transition from conventional propagation models toward learning-based methods capable of exploiting large-scale measurement datasets and geographical information.

A. Traditional Propagation Models

Early wireless coverage prediction relied primarily on empirical propagation models derived from extensive measurement campaigns. Among the most widely adopted models are the Okumura model and its analytical extension, the Hata model, which provide path-loss estimations for urban, suburban, and rural environments. Subsequent developments such as the COST-231 Hata model extended the applicable frequency range to cellular systems operating above 1 GHz, while the COST-231 Walfisch-Ikegami model incorporated the influence of dense urban morphology and non-line-of-sight (NLOS) propagation conditions [1]–[3].

Other commonly used models include the Irregular Terrain Model (ITM), also known as the Longley-Rice model, and the ECC-33 model. These approaches integrate terrain

characteristics and diffraction effects to improve prediction accuracy over irregular landscapes [4], [5]. Despite their simplicity and low computational requirements, empirical models often exhibit significant prediction errors when applied outside the environments for which they were originally calibrated. Previous studies have reported prediction errors ranging from 10 dB to more than 40 dB in complex environments due to the inability of these models to capture local terrain morphology, vegetation, and urban obstructions [6], [10].

Several investigations have specifically evaluated empirical models for LPWAN technologies such as LoRa and LoRaWAN. Petäjärvi et al. [7] demonstrated that LoRa coverage can exceed 15 km under favorable conditions, although the accuracy of conventional path-loss models deteriorates at long distances. El Chall et al. [8] proposed environment-specific propagation models for LoRaWAN deployments in Lebanon and showed that customized models outperform standard ITU-R, Okumura-Hata, and COST-231 formulations. Similar conclusions were reached by Callebaut and Van der Perre [9], who observed significant propagation variability across urban, forest, and coastal environments and highlighted the limitations of traditional empirical formulations for LoRa networks.

B. Terrain-Aware and Measurement-Based Coverage Modeling

To improve prediction accuracy, several researchers have incorporated terrain information and geographic data into propagation modeling. Demetri et al. [10] proposed an automated LoRa coverage estimation framework combining remote sensing information with propagation modeling, reducing prediction errors compared with conventional approaches. Likewise, Bezerra et al. [11] evaluated ITM, ITWOM, and Okumura-Hata predictions against real LoRaWAN measurements and demonstrated that no single propagation model consistently performs well across all deployment scenarios.

Additional studies emphasized the importance of environmental characteristics such as terrain elevation, vegetation density, weather conditions, and Fresnel zone obstruction. Elijah et al. [13] identified terrain morphology as one of the dominant factors affecting LoRa communication performance, while Bezerra et al. [12] reported measurable signal degradation caused by seasonal weather variations. Lima et al. [14] further developed dedicated propagation models for LoRa deployments in Amazonian regions and demonstrated superior performance compared with conventional Okumura-Hata formulations.

Although terrain-aware empirical models generally outperform purely distance-based approaches, their performance remains strongly dependent on calibration procedures and environmental assumptions. Consequently, the development of data-driven techniques capable of automatically learning complex propagation patterns has attracted increasing research attention.

C. Machine Learning for Coverage and Path-Loss Prediction

Recent advances in machine learning have significantly transformed wireless coverage prediction. Machine learning

algorithms can automatically discover nonlinear relationships between signal measurements, network parameters, and environmental features, often achieving substantially higher accuracy than conventional propagation models [15], [17].

Several studies have investigated machine learning for path-loss estimation and wireless coverage mapping. Alimpertis et al. [16] generated city-scale signal-strength maps using Random Forest regression and demonstrated substantial improvements over traditional interpolation methods. Moraitis et al. [18] compared multiple machine learning techniques for rural path-loss prediction and showed that ensemble-based methods consistently outperform conventional regression approaches.

For cellular networks, Fauzi et al. [19] evaluated Linear Regression, Artificial Neural Networks, Support Vector Machines, Random Forests, and Gaussian Process Regression for RSRP prediction. Their results demonstrated that machine learning models significantly outperform conventional propagation models, with Random Forest and Gaussian Process Regression providing the most favorable trade-offs between prediction accuracy and computational complexity.

Similarly, He et al. [20] proposed a Random Forest-based path-loss prediction framework capable of capturing nonlinear propagation effects in mobile communication systems. The Machine Learning-Based Online Coverage Estimator (MLOE) introduced by Fauzi et al. [21] further demonstrated that machine learning models can be integrated into practical network-planning tools while maintaining high prediction accuracy and fast inference times.

Within the LPWAN domain, Hosseinzadeh et al. [22] proposed an Adaptive Neuro-Fuzzy Inference System (ANFIS) for LoRaWAN propagation modeling, showing improved adaptability compared with conventional empirical approaches. More recently, Hosseinzadeh et al. [23] investigated explainable machine learning techniques for LoRaWAN link-budget analysis and demonstrated that ensemble learning algorithms such as Random Forests and Gradient Boosting achieve robust predictive performance while providing insights into feature importance. González-Palacio et al. [24] also employed machine learning techniques to jointly model path loss and shadow fading for LoRaWAN systems, achieving substantial gains in network energy efficiency.

D. Deep Learning and Hybrid Learning Approaches

Beyond conventional machine learning algorithms, deep learning approaches have recently emerged as promising solutions for wireless coverage estimation. Mobile network coverage prediction based on multi-modal Deep Neural Networks and semantic segmentation demonstrated that image-based environmental representations can significantly improve prediction accuracy without relying on explicit propagation equations [25]. However, these methods typically require large training datasets, extensive computational resources, and complex model architectures.

Hybrid approaches combining empirical propagation models with machine learning have also been proposed. Alobaidy et al. [26] introduced a family of stacked semi-empirical machine learning models for LoRa propagation characterization

in tropical environments. Their framework combines Free-Space Path Loss calculations with Support Vector Machines, Stepwise Regression, Ensemble Bagged Trees, and Artificial Neural Networks. Using more than 43,000 real-world measurements, the proposed models achieved prediction accuracies up to $R^2 = 0.89$ while maintaining high inference speed. Their results confirmed that hybrid machine learning models can substantially outperform conventional propagation models such as ITM, COST-231 Hata, and ECC-33.

The reviewed literature demonstrates the evolution of wireless coverage prediction from traditional empirical propagation models toward data-driven machine learning and deep learning approaches. While significant improvements in prediction accuracy have been reported, existing studies differ considerably in terms of modeling methodology, input features, computational complexity, and deployment objectives.

To provide a clearer overview of the state of the art and identify the remaining challenges, Table I summarizes representative studies, highlighting their main advantages and limitations.

E. Research Gap and Contribution

As shown in Table I, empirical propagation models remain attractive because of their simplicity and low computational requirements. However, their performance is highly dependent on the deployment environment and often degrades in heterogeneous terrains. Spatial interpolation techniques can improve local coverage estimation accuracy but require dense measurement campaigns and generally exhibit limited extrapolation capability.

Recent machine learning and deep learning approaches have demonstrated superior prediction accuracy by exploiting measurement data and environmental information. Nevertheless, several limitations remain. Many studies focus on cellular networks rather than LPWAN technologies, while others rely on computationally intensive architectures, large training datasets, or environment-specific calibration procedures. Furthermore, the combined exploitation of terrain morphology, land-use information, obstruction descriptors, Fresnel-zone characteristics, and neighboring signal statistics remains relatively limited in LoRaWAN coverage prediction.

To address these limitations, this work proposes a machine learning-based framework for wireless coverage estimation using real-world LoRaWAN measurements and GIS-derived features. The proposed approach integrates radio transmission parameters, geometric descriptors, terrain elevation statistics, obstruction and Fresnel-zone metrics, land-use information, and neighboring signal features within an Extra Trees regression framework. Trained on more than 10,000 RSSI measurements collected in Da Nang and Hai Phong, Vietnam, the framework aims to provide accurate RSSI prediction while remaining suitable for automated wireless coverage mapping and network planning applications.

III. METHODOLOGY AND PROPOSED METHOD

A. Dataset

The objective of this work is to estimate the received LoRaWAN received signal strength indicator (RSSI) at arbitrary

TABLE I
COMPARISON OF EXISTING WIRELESS COVERAGE PREDICTION APPROACHES AND IDENTIFIED RESEARCH GAPS

Study	Method	Advantages	Limitations / Research Gap
Okumura-Hata, COST-231, ECC-33, ITM [1]–[4]	Empirical propagation models	Simple implementation, low computational complexity, widely used in network planning	Require environment-specific calibration and often fail to accurately capture local terrain, vegetation, and urban obstructions. Prediction accuracy degrades in heterogeneous environments.
Demetri et al. [10]	Remote sensing-assisted LoRa coverage estimation	Integrates geographical information and reduces prediction error compared with conventional models	Still relies on propagation modeling assumptions and limited environmental adaptability.
Bezerra et al. [11]	ITM, ITWOM and Okumura-Hata evaluation	Demonstrates the importance of terrain-aware propagation analysis	No single propagation model performs consistently well across different deployment scenarios.
El Chall et al. [8]	Environment-specific LoRaWAN propagation models	Improved prediction accuracy over standard empirical models	Models remain location-dependent and require dedicated calibration campaigns.
Lima et al. [14]	LoRa propagation models for Amazon regions	Improved performance in dense vegetation environments	Limited generalization to different geographic and urban environments.
Alimpertis et al. [16]	Random Forest signal strength mapping	Demonstrates effectiveness of machine learning for large-scale coverage prediction	Focuses primarily on citywide cellular signal mapping and does not exploit detailed terrain descriptors.
Fauzi et al. [19]	LR, ANN, SVM, RF, and GPR	Comprehensive comparison of supervised ML algorithms for RSRP prediction	Primarily focused on cellular networks rather than LPWAN technologies.
MLOE [21]	Random Forest online coverage estimator	Accurate and computationally efficient deployment framework	Uses a limited set of geographical and environmental features.
Hosseinzadeh et al. [22]	ANFIS-based LoRaWAN propagation model	Captures nonlinear propagation effects and improves adaptability	Model complexity increases with feature dimensionality and training data requirements.
Hosseinzadeh et al. [23]	Explainable ML for LoRaWAN modeling	Provides both prediction capability and feature interpretability	Focused mainly on link-budget analysis rather than large-scale coverage map generation.
González-Palacio et al. [24]	ML-based path-loss and shadowing estimation	Improves LoRaWAN energy efficiency and propagation prediction	Primarily designed for transmission power optimization.
Zeng et al. [25]	Deep Neural Network + Semantic Segmentation	High prediction accuracy using environmental image information	Requires large datasets, high computational resources, and complex training procedures.
Alobaidy et al. [26]	Hybrid stacked ML models for LoRa propagation	High prediction accuracy and strong generalization across tropical environments	Requires multiple stacked models and increased model complexity.
This Work	Extra Trees-based RSSI prediction using GIS and spatial features	Integrates transmission parameters, terrain descriptors, land-use information, Fresnel-zone metrics, obstruction features, and neighboring RSSI statistics into a unified machine-learning framework for accurate LoRaWAN coverage prediction.	Performance depends on the availability of representative reference measurements, while GIS-based feature extraction dominates the prediction time.

geographical locations using a supervised machine learning approach.

The dataset consists of LoRaWAN measurements collected in two Vietnamese urban environments, Da Nang and Hai Phong, using 13 gateways operating under different radio configurations. Each measurement includes the transmitter location, the associated gateway, the operating frequency, the spreading factor, and the measured RSSI.

To enrich the prediction model with environmental information, external geographic datasets were also used during feature engineering. Terrain elevation data were obtained from the NASA Shuttle Radar Topography Mission (SRTM), while land-use information was extracted from OpenStreetMap.

Several preprocessing operations were applied before model training, including data parsing, duplicate removal, and con-

sistency checks. During exploratory analysis, a small number of duplicated measurements collected at identical locations exhibited physically inconsistent RSSI values. Rather than discarding all duplicated observations, only the most extreme outliers were removed, preserving the remaining measurements and the natural variability of wireless propagation.

The final cleaned dataset contains 10,203 measurements. These measurements constitute the reference dataset from which the geometric, terrain-related, and nearest-neighbor features described in the Section III-C are generated.

B. Overall System Architecture

The proposed framework is organized into three main stages: data acquisition, offline model training, and online prediction and coverage estimation.

Figure 1 presents the overall architecture of the proposed RSSI prediction system.

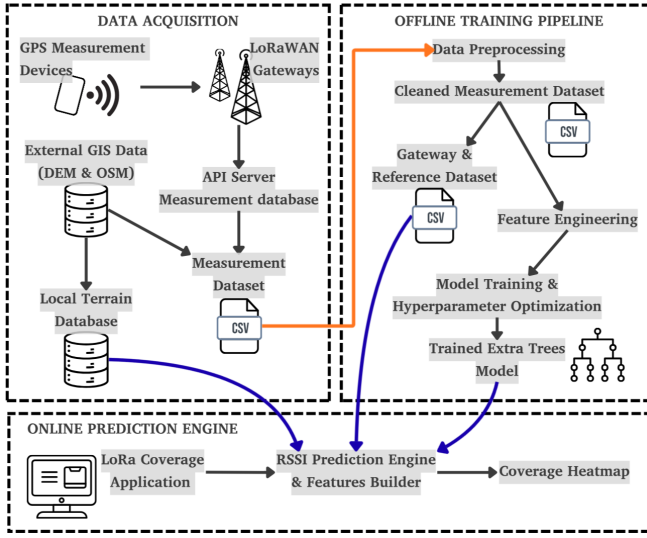


Fig. 1. Overall architecture of the proposed LoRaWAN RSSI prediction framework.

During the data acquisition stage, wireless measurements are collected through LoRaWAN devices and gateways and retrieved from the measurement database through an API. In parallel, geographic information including Digital Elevation Models (DEM) and OpenStreetMap (OSM) data is obtained from external sources. These datasets are combined to create the measurement dataset used throughout the development process. In addition, terrain-related information is stored in a local terrain database for efficient reuse during prediction.

The offline training stage transforms the collected measurements into engineered features and uses them to train and optimize the machine learning model. The resulting artifacts include the trained Extra Trees model as well as the gateway and reference measurement database required during deployment.

The online prediction stage is implemented through a dedicated prediction engine. Given a query location and radio configuration, the engine loads the trained model together with the gateway database, reference measurements, and local terrain database. The required features are then generated dynamically, including geometric, terrain, and nearest-neighbor descriptors, before estimating the RSSI value. The predicted signal strengths can subsequently be used to generate wireless coverage maps over arbitrary geographical areas.

For simplicity, Figure 1 does not explicitly show the use of the local terrain database during feature engineering. In practice, terrain information extracted from DEM and OSM data is incorporated into the measurement dataset during feature generation in the offline training stage, while the same local terrain database is reused by the prediction engine during online inference.

C. Feature Engineering

The predictive variables were designed to capture three complementary aspects of radio propagation: radio configu-

ration, geometric and environmental characteristics, and local spatial information extracted from neighboring measurements.

1) *Radio Features*: Only parameters available before transmission are used, namely the operating frequency and the spreading factor. Bandwidth was excluded since it remained constant throughout the dataset.

2) *Geometric Features*: The relative geometry between the gateway and the receiver is represented using the three-dimensional distance, the logarithm of the 3D distance, the latitude and longitude offsets, the azimuth angle, the gateway and receiver elevations, the elevation difference, and the elevation angle.

The three-dimensional propagation distance is computed as

$$d_{3D} = \sqrt{d_{2D}^2 + (\Delta h)^2},$$

where d_{2D} is obtained from the Haversine formula and Δh is the elevation difference between both endpoints.

The logarithmic distance is also included since signal attenuation approximately follows a logarithmic law.

3) *Terrain Features*: Terrain elevation information was extracted from the NASA Shuttle Radar Topography Mission (SRTM) digital elevation model [27], while land-use descriptors were obtained from OpenStreetMap.

For each gateway–receiver path, the terrain profile is sampled and summarized through several descriptors, including the mean, standard deviation, minimum, maximum, and range of terrain elevation, the average terrain slope, terrain roughness, maximum obstruction, Fresnel obstruction ratio, minimum and mean Fresnel clearance, and the residential land-use ratio.

These features provide a simplified representation of diffraction, shadowing and environmental attenuation effects.

a) *Terrain Descriptor Computation*: The local terrain slope is estimated from a 3×3 DEM window centered on the receiver location. Let g_x and g_y denote the horizontal and vertical elevation gradients obtained through finite differences. The slope descriptor is computed as

$$S = \frac{1}{N} \sum_{i=1}^N \sqrt{g_{x,i}^2 + g_{y,i}^2}.$$

Terrain roughness is defined as the standard deviation of elevations within a local 5×5 DEM neighborhood:

$$R = \sqrt{\frac{1}{N} \sum_{i=1}^N (h_i - \bar{h})^2}.$$

Higher values indicate more irregular local terrain conditions.

b) *Line-of-Sight and Obstruction Features*: For each gateway–receiver path, terrain elevations are sampled every 30 m along the propagation path.

The line-of-sight (LOS) height at sample position i is computed through linear interpolation between the gateway and receiver antenna heights:

$$h_{\text{LOS}}(i) = h_g + \frac{i}{N-1} (h_r - h_g).$$

The obstruction at position i is then defined as

$$O_i = h_{\text{terrain}}(i) - h_{\text{LOS}}(i).$$

The maximum obstruction feature corresponds to

$$O_{\max} = \max_i O_i.$$

Positive values indicate terrain penetrating the direct propagation path.

The first Fresnel zone radius at a given point along the path is computed as

$$F_1 = \sqrt{\frac{\lambda d_1 d_2}{d_1 + d_2}},$$

where λ is the wavelength and d_1 and d_2 are the distances from the considered point to the transmitter and receiver, respectively.

The Fresnel clearance is defined as

$$C_i = h_{\text{LOS}}(i) - F_1(i) - h_{\text{terrain}}(i).$$

The Fresnel obstruction ratio corresponds to the proportion of sampled points for which

$$C_i < 0.$$

This descriptor quantifies the fraction of the propagation path affected by Fresnel-zone blockage.

D. Nearest-Neighbor Spatial Features

A major contribution of the proposed method is the integration of local historical measurements into the prediction process.

Rather than directly using K-nearest-neighbor interpolation as the final predictor, neighboring observations are transformed into statistical descriptors which become additional input variables for the machine learning model.

1) *Gateway-Specific Spatial Indexing*: Historical measurements are first grouped according to their associated gateway.

For each gateway, measurement coordinates are projected into a local Cartesian system:

$$x = 111000 \times \text{latitude},$$

$$y = 111000 \cos(\bar{\phi}) \times \text{longitude},$$

where $\bar{\phi}$ denotes the average latitude of the dataset.

A dedicated KD-Tree is then constructed for each gateway, allowing efficient nearest-neighbor retrieval with logarithmic complexity.

2) *Neighbor Selection*: Given a query point, the corresponding gateway KD-Tree is first queried to retrieve a set of candidate neighbors.

Measurements located within a minimum radius

$$d < d_{\min}$$

are discarded in order to avoid duplicated samples collected at the same physical position.

The remaining candidates are ranked according to a composite score:

$$S_i = d_i + \lambda |D_i - D_q|,$$

where d_i denotes the spatial distance to neighbor i , D_i is the gateway distance of neighbor i , D_q is the gateway distance of the query point, and λ is a weighting coefficient.

This criterion favors neighboring measurements that are geographically close to the query location while exhibiting a gateway distance similar to that of the query point.

The neighbor selection parameters were treated as hyper-parameters and optimized empirically during model development.

The final configuration was

$$K = 9,$$

$$K_{\text{search}} = 11,$$

$$d_{\min} = 0.1 \text{ m},$$

$$\lambda = 1.1.$$

The parameter K determines the number of neighbors retained for feature generation, whereas K_{search} specifies the number of candidate neighbors initially retrieved from the KD-Tree.

A larger value of K_{search} provides additional flexibility during the ranking stage by allowing the algorithm to select the most relevant neighbors according to the composite score of Eq. (XX). The final values were obtained through empirical optimization.

3) *Neighbor Statistics*: After selecting the best neighbors, several statistical descriptors are computed, including the RSSI, spatial distance, and gateway distance of the closest neighbor, the mean and weighted mean neighbor RSSI, the neighbor RSSI standard deviation, as well as the mean spatial and gateway distances.

The weighted RSSI average is defined as

$$\bar{R} = \frac{\sum_i R_i e^{-d_i/\tau}}{\sum_i e^{-d_i/\tau}},$$

where R_i is the RSSI of neighbor i and d_i its spatial distance. τ is a distance decay parameter controlling the influence of neighboring observations. In this work, $\tau = 30$ m was selected empirically.

Algorithm 1 Training procedure**Require:** Measurement dataset $\mathcal{D}$

```

0: for each cross-validation fold do
0:   Split  $\mathcal{D}$  into training and validation sets
0:   Build gateway-specific KD-Trees using training sam-
     ples only
0:   Generate neighbor features for training samples
0:   Generate neighbor features for validation samples using
     the training KD-Trees
0:   Train Extra Trees model
0:   Evaluate  $R^2$ , MAE and RMSE
0: end for
0: Retrain final model on the complete dataset
0: Build final gateway-specific KD-Trees
0: Save model and auxiliary databases
  =0

```

Unlike conventional KNN regression, these neighboring measurements are not directly used as predictions. Instead, they enrich the feature space and allow the learning algorithm to model complex nonlinear interactions between local propagation behavior and environmental conditions.

E. Machine Learning Model

Algorithm 1 summarizes the training and evaluation procedure used throughout this work.

Several ensemble regression algorithms were investigated during development. Among them, the Extra Trees Regressor provided the best compromise between predictive performance and computational efficiency.

Hyperparameter optimization was performed using grid search combined with shuffled five-fold cross-validation.

The final Extra Trees configuration consists of 650 trees, a maximum tree depth of 18, a maximum feature ratio of 0.7, and a minimum of 10 samples required for each split.

Larger forests containing up to 1500 trees were also evaluated but provided negligible performance improvements while significantly increasing inference cost.

F. Model Evaluation

Model performance was assessed using shuffled five-fold cross-validation.

At each iteration, approximately 80% of the dataset was used for training and the remaining 20% for validation.

Three standard regression metrics were considered: the coefficient of determination (R^2), the mean absolute error (MAE), and the root mean squared error (RMSE).

The final model achieved:

$$R^2 = 0.9285 \pm 0.0065,$$

$$MAE = 1.93 \text{ dBm},$$

$$RMSE = 2.92 \text{ dBm}.$$

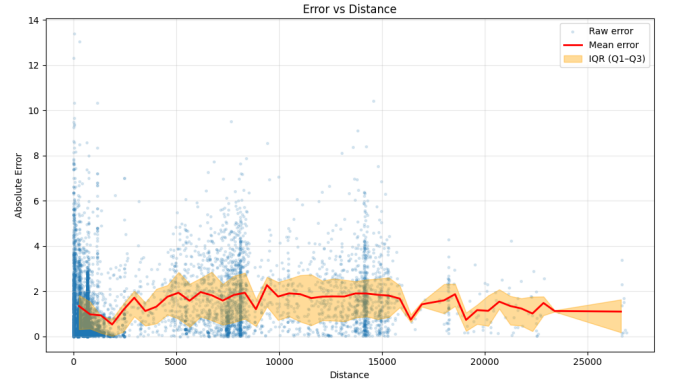


Fig. 2. Absolute prediction error as a function of transmitter–gateway distance. The curve shows median error and interquartile range per distance bin.

These results demonstrate that combining physical propagation descriptors with local historical observations significantly improves RSSI prediction accuracy.

Figure 2 shows the evolution of the absolute prediction error as a function of the transmitter–gateway distance. For each distance bin, the median error and interquartile range are reported.

Overall, the results indicate that the prediction error does not systematically increase with distance. Although the dataset contains a higher density of samples at short ranges due to the measurement campaign, the model maintains stable performance across different distance regimes. This suggests that the learned representation generalizes beyond local spatial neighborhoods and is not solely driven by proximity effects in the training data.

Slight increases in variability can be observed in specific distance intervals, which may be associated with heterogeneous propagation conditions and local measurement inconsistencies rather than a monotonic degradation of model performance with distance.

G. Inference Engine

To facilitate practical deployment, the prediction engine was designed independently from the training framework. The deployed system only requires the artifacts generated during the offline training stage and never performs model retraining or dataset reconstruction.

At initialization, the inference engine loads the trained Extra Trees model together with the gateway database, the reference measurement database, the gateway-specific KD-Trees, and the terrain and land-use datasets.

These data structures are loaded only once and remain resident in memory during operation.

Algorithm 2 summarizes the online prediction procedure.

The deployment strategy includes several practical optimizations intended to improve prediction efficiency and robustness.

If no gateway is specified, the engine automatically selects one according to the deployment strategy implemented by the application. Furthermore, when a reference measurement

Algorithm 2 Online RSSI prediction

Require: Query location (lat, lon)

```

0: if gateway not specified then
0:   Select gateway using the deployment strategy
0: end if
0: if reference measurement exists within 30 m then
0:   return measured RSSI
0: end if
0: if query location is outside the gateway operating range
   then
0:   return -120 dBm
0: end if
0: Compute geometric descriptors
0: Extract terrain descriptors
0: Retrieve neighboring measurements
0: Compute neighbor statistics
0: Assemble feature vector  $\mathbf{x}$ 
0: Predict RSSI using the trained Extra Trees model
   return RSSI estimate
=0

```

is available within a predefined distance threshold, the corresponding measured RSSI is returned directly instead of invoking the machine learning model. Similarly, predictions requested outside the observed operating range of a gateway are rejected by returning a default low RSSI value.

These optimizations are applied only during deployment and are independent of the machine learning model itself. They do not affect the training procedure nor the evaluation protocol presented in this work.

This modular architecture enables efficient integration into interactive wireless coverage mapping applications where large numbers of prediction queries must be processed.

IV. EXPERIMENTAL SETUP

A. Dataset Description

The dataset used in this study contains 10,203 cleaned LoRaWAN measurements collected in two Vietnamese cities: Da Nang and Hai Phong.

Measurements were obtained from 13 LoRaWAN gateways and three end devices (board01, node01, and node3). Each record contains the geographical coordinates of the transmission point, gateway information, radio configuration parameters, and the measured RSSI value.

After the initial preprocessing stage, a small number of physically inconsistent duplicate measurements were identified. Only the most extreme outliers were removed while preserving the remaining observations.

The final dataset contains 10,203 measurements, including 8,698 collected in Da Nang and 1,531 in Hai Phong, acquired using 13 gateways and three end devices.

Table II summarizes the RSSI distribution.

B. Radio Configuration Distribution

The dataset includes three spreading factors: SF7, SF10, and SF12.

TABLE II
RSSI STATISTICS OF THE FINAL DATASET

Statistic	RSSI (dBm)
Minimum	-135
25th percentile	-114
Median	-110
Mean	-108.44
75th percentile	-105
Maximum	-28
Standard deviation	10.96

TABLE III
DISTRIBUTION OF SPREADING FACTORS

Spreading Factor	Percentage (%)
SF7	27.72
SF10	24.45
SF12	47.83

Measurements were collected using several frequencies within the Vietnamese LoRaWAN band.

TABLE IV
FREQUENCY DISTRIBUTION

Frequency (MHz)	Percentage (%)
921.4	43.84
921.6	3.09
921.8	1.08
922.0	0.98
922.2	48.79
922.4	0.90
922.6	0.73
922.8	0.60

C. Cross-Validation Protocol

Model evaluation was performed using shuffled five-fold cross-validation.

At each iteration, approximately 80% of the data were used for training and 20% for validation. The folds were randomly shuffled before splitting.

To prevent information leakage, the reference measurement database used for nearest-neighbor feature generation was reconstructed independently for each cross-validation fold. Neighbor descriptors for validation samples were therefore computed exclusively from measurements belonging to the corresponding training partition.

This protocol was selected to evaluate the general predictive performance of the proposed model while maximizing the utilization of the available dataset.

D. Model Configuration

The final prediction model is based on the Extra Trees Regressor algorithm.

After hyperparameter optimization, the final Extra Trees model was configured with 650 trees, a maximum depth of 18, a minimum of 10 samples required for each split, and a maximum feature ratio of 0.7.

The nearest-neighbor feature generation module was configured with $K = 9$, $K_{search} = 11$, a minimum distance

threshold of 0.1 m, and a gateway-distance weighting coefficient $\lambda = 1.1$.

This configuration provided the best trade-off between accuracy and computational complexity.

E. Evaluation Metrics

The prediction performance was evaluated using three standard regression metrics: the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

The coefficient of determination is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

where y_i is the measured RSSI, $\hat{y}_i$ is the predicted RSSI, and $\bar{y}$ is the average RSSI.

The Mean Absolute Error is given by:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

while the Root Mean Square Error is defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

MAE measures the average prediction error in dBm, while RMSE penalizes larger prediction errors more strongly. Together with R^2 , these metrics provide a comprehensive evaluation of model performance.

V. RESULTS AND DISCUSSION

A. Performance of Machine Learning Models

Several supervised learning algorithms were evaluated during the development process in order to identify the most suitable model for RSSI prediction. Hyperparameter optimization was performed independently for each algorithm using five-fold cross-validation.

Table VI summarizes the obtained results.

TABLE V
PERFORMANCE COMPARISON OF EVALUATED MACHINE LEARNING MODELS

Model	MAE (dBm)	RMSE (dBm)	R^2
SVR	2.79	5.49	0.7610
MLP Regressor	2.58	4.19	0.8609
HistGradientBoosting	2.22	3.72	0.8904
XGBoost	2.14	3.80	0.8857
Random Forest	2.11	3.42	0.9073
Extra Trees	2.03	3.45	0.9056
Final Proposed Extra Trees	1.93	2.92	0.9285

Among the evaluated algorithms, tree-based ensemble methods consistently achieved the best results. Extra Trees and Random Forest clearly outperformed neural-network and kernel-based approaches, suggesting that the relationship between RSSI and environmental descriptors is highly nonlinear and benefits from ensemble partitioning strategies.

After the introduction of terrain-aware and nearest-neighbor features, a detailed comparison was conducted between Random Forest and Extra Trees regressors using identical feature sets and five-fold cross-validation.

Table VI summarizes the obtained results.

TABLE VI
PERFORMANCE COMPARISON OF TREE-BASED ENSEMBLE MODELS

Model	MAE (dBm)	RMSE (dBm)	R^2
Random Forest	2.05	3.24	0.9117
Extra Trees	2.00	3.21	0.9135
Final Proposed Extra Trees	1.93	2.92	0.9285

The results indicate that Extra Trees consistently achieved slightly better prediction accuracy than Random Forest. Although the difference between both ensemble methods remained moderate, Extra Trees provided lower MAE and RMSE values while obtaining the highest coefficient of determination.

This behavior can be explained by the stronger randomization introduced during tree construction, which reduces variance and improves generalization on heterogeneous propagation environments. Consequently, Extra Trees was selected as the baseline model for further optimization.

The final performance improvement was obtained through additional feature engineering, nearest-neighbor parameter tuning, and the removal of a small number of physically inconsistent duplicate measurements. The resulting model achieved an average cross-validation performance of $R^2 = 0.9285 \pm 0.0065$, MAE = 1.93 dBm, and RMSE = 2.92 dBm.

Figure 3 compares the predicted RSSI values with the corresponding ground-truth measurements. Most samples are concentrated around the identity line, indicating good agreement between predictions and observations over the full RSSI range.

B. Impact of Feature Engineering

A progressive feature engineering strategy was adopted throughout the development process. Initial experiments relied exclusively on radio parameters and terrain-derived descriptors. Subsequently, spatial nearest-neighbor features extracted from previously observed measurements were introduced.

The addition of neighbor-based RSSI descriptors produced the largest performance improvement. In particular, the weighted mean RSSI of neighboring measurements and the RSSI of the closest reference point emerged as highly informative predictors.

Furthermore, several candidate features were evaluated and removed when they failed to improve cross-validation performance. Examples include path-based neighbor descriptors and redundant geometric features exhibiting strong correlation with existing variables.

The final feature set combines radio transmission parameters, geometric descriptors, terrain characteristics, Fresnel-zone indicators, land-use information, and spatial neighbor statistics, resulting in the highest observed prediction accuracy.

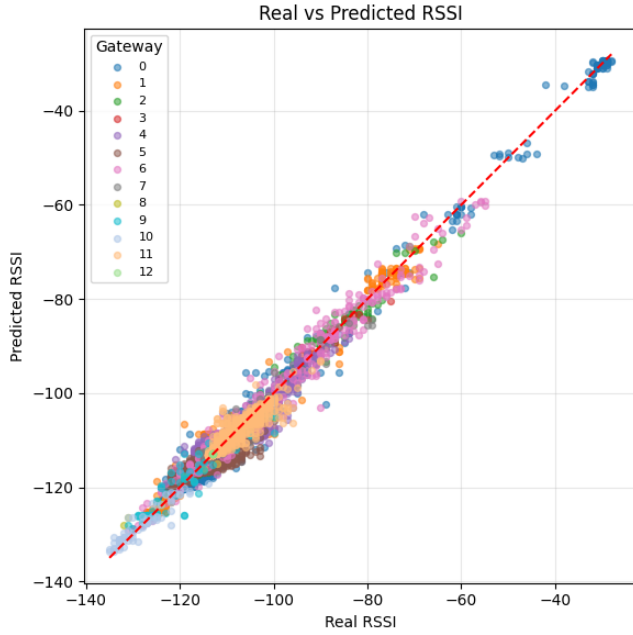


Fig. 3. Predicted RSSI versus measured RSSI. The dashed line represents the ideal prediction ($y = x$).

C. Feature Importance Analysis

Feature importance was evaluated using the impurity-based importance scores provided by the final Extra Trees model.

Table VII presents the most influential variables.

TABLE VII
TOP 10 MOST IMPORTANT FEATURES

Feature	Importance
Residential ratio	0.159
Gateway ID	0.119
Neighbor RSSI weighted mean	0.083
Frequency	0.079
Neighbor RSSI mean	0.061
RSSI closest point	0.059
Elevation angle	0.047
Spreading factor	0.034
Log distance (3D)	0.034
Elevation	0.031

The results indicate that both environmental and spatial-context features play a major role in prediction accuracy. Neighbor-based RSSI descriptors rank among the most influential variables, confirming that local propagation behavior can be effectively captured using nearby measurements.

Terrain-related features such as elevation angle, terrain statistics, slope, and roughness also contribute significantly, highlighting the importance of incorporating geographic information into wireless coverage prediction models.

Although the residential land-use ratio obtained the highest impurity-based importance score, this does not necessarily imply a direct causal relationship with RSSI. Tree-based importance measures tend to favor variables involved in many decision splits and should therefore be interpreted qualitatively rather than quantitatively.

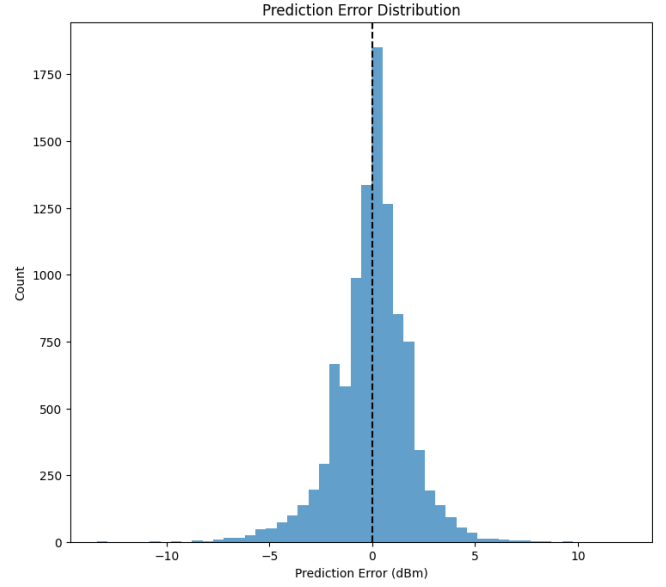


Fig. 4. Distribution of prediction errors on the evaluation dataset.

Interestingly, Fresnel-zone and obstruction-related features exhibit relatively low individual importance. Nevertheless, removing these variables resulted in a measurable performance degradation during feature selection experiments, suggesting that they provide complementary information to the model.

D. Error Analysis

Inspection of the largest residuals revealed that several correspond to duplicated measurements acquired at the same geographical location but associated with substantially different RSSI values.

For example, measurements recorded approximately 24.5 meters from a gateway showed RSSI values ranging from -114 dBm to -84 dBm. Similarly, another location situated approximately 2457 meters from its gateway presented RSSI values between -112 dBm and -90 dBm.

Such variations are difficult to justify from a propagation perspective and are likely caused by temporary interference, corrupted packets, hardware instability, or exceptional environmental conditions.

Only the most extreme outliers were removed while preserving the remaining observations.

This targeted cleaning procedure improved the model performance from approximately:

$$R^2 = 0.9257$$

to

$$R^2 = 0.9285$$

while maintaining the integrity of the original dataset.

The maximum remaining prediction error is approximately 13 dB, with the vast majority of predictions exhibiting significantly lower deviations. Figure 4 shows the distribution of prediction errors.

TABLE VIII
PERFORMANCE COMPARISON WITH REPRESENTATIVE MACHINE LEARNING-BASED WIRELESS COVERAGE PREDICTION STUDIES

Reference	Technology	Model	RMSE (dB)	R^2	Main Features
[23]	LoRaWAN	Random Forest	5.70	0.77	Link-budget and propagation variables
[26]	LoRa	Hybrid ML (M1)	4.77	0.89	Propagation, terrain and channel descriptors
[21]	LTE	Random Forest	2.65	0.93	Geometric and radio-network features
[25]	Cellular	DNN + Semantic Segmentation	1.97	–	Numerical + image-derived environmental features
Proposed Method	LoRaWAN	Extra Trees	2.92	0.9285	Terrain, Fresnel, land use, neighbor RSSI

TABLE IX
METHODOLOGICAL DIFFERENCES AFFECTING DIRECT COMPARISON

Reference	Evaluation Context	Remarks for Comparison
[23]	LoRaWAN path-loss modelling	Focuses primarily on propagation modelling and link-budget analysis. Uses fewer environmental descriptors than the proposed approach.
[26]	LoRa path-loss prediction	Employs a hybrid stacked machine learning architecture and evaluates path-loss prediction rather than RSSI coverage mapping. Direct comparison is informative but not strictly equivalent.
[21]	LTE/4G coverage estimation	Uses cellular-network measurements and engineered radio-network features. Different radio technology and propagation characteristics from LoRaWAN.
[25]	Cellular coverage prediction	Relies on semantic-segmentation-derived environmental representations and deep neural networks. Requires image-based feature extraction and significantly higher model complexity.
Proposed Method	LoRaWAN RSSI prediction	Combines GIS-derived descriptors and neighboring RSSI statistics using an Extra Trees ensemble regressor.

E. Comparison with Existing Methods

To position the proposed approach with respect to recent machine learning-based wireless coverage prediction studies, Table VIII compares the predictive performance reported in representative works. Since different studies employ different radio technologies, datasets, feature engineering strategies, and evaluation protocols, the comparison should be interpreted as an indication of relative performance rather than a strict benchmark.

As shown in Table VIII, the proposed model achieves competitive performance with an RMSE of 2.92 dB and an R^2 of 0.9285. Among LoRa-oriented studies, the proposed approach substantially improves upon the Random Forest model reported in [23] and also outperforms the best hybrid model reported in [26], reducing the prediction error from 4.77 dB to 2.92 dB while achieving a higher coefficient of determination.

The proposed model also approaches the performance of MLOE [21], which reports an RMSE of 2.65 dB and an R^2 of 0.93. However, MLOE targets LTE coverage prediction and relies on cellular-network-specific features, making direct comparison difficult. Similarly, the DNN-SS framework [25] achieves a lower RMSE of 1.97 dB, but uses a considerably more complex deep-learning architecture incorporating semantic segmentation and image-derived environmental representations.

Table IX summarizes the principal methodological differences between the studies. Unlike deep-learning-based approaches, the proposed framework relies exclusively on engineered GIS-derived features, Fresnel-zone descriptors, land-use information, and neighboring RSSI statistics. The resulting Extra Trees model therefore offers an attractive trade-off between prediction accuracy, computational simplicity, and

scalability for large-area LoRaWAN coverage map generation.

F. Independent Field Validation

To further assess the proposed framework, an independent measurement campaign was conducted using approximately 800 LoRaWAN measurements that were not included in either the training or cross-validation datasets.

Unlike the original dataset, a large proportion of these measurements (approximately 600 samples) were received by a previously unseen gateway that was not available in the reference gateway database used during training. Consequently, the prediction engine could not directly use the actual receiving gateway.

Instead, when no matching gateway was found, the inference engine selected the gateway associated with the geographically closest reference measurements. For the new deployment, this substitute gateway was located approximately 190 m from the actual gateway. As a result, all geometry-dependent and terrain-related descriptors, including propagation distance, elevation profile, obstruction metrics, and Fresnel-zone features, were computed with respect to this surrogate gateway rather than the true receiving gateway.

Under these conditions, the proposed framework achieved a Mean Absolute Error (MAE) of 3.31 dBm and a Root Mean Square Error (RMSE) of 4.60 dBm.

The prediction error is therefore higher than that obtained during cross-validation (MAE = 1.93 dBm and RMSE = 2.92 dBm). This degradation is expected since a substantial fraction of the test measurements relies on an approximate gateway location rather than the true receiving gateway. Despite this mismatch, the model maintains a prediction error close to 3 dBm on average, indicating that the proposed feature

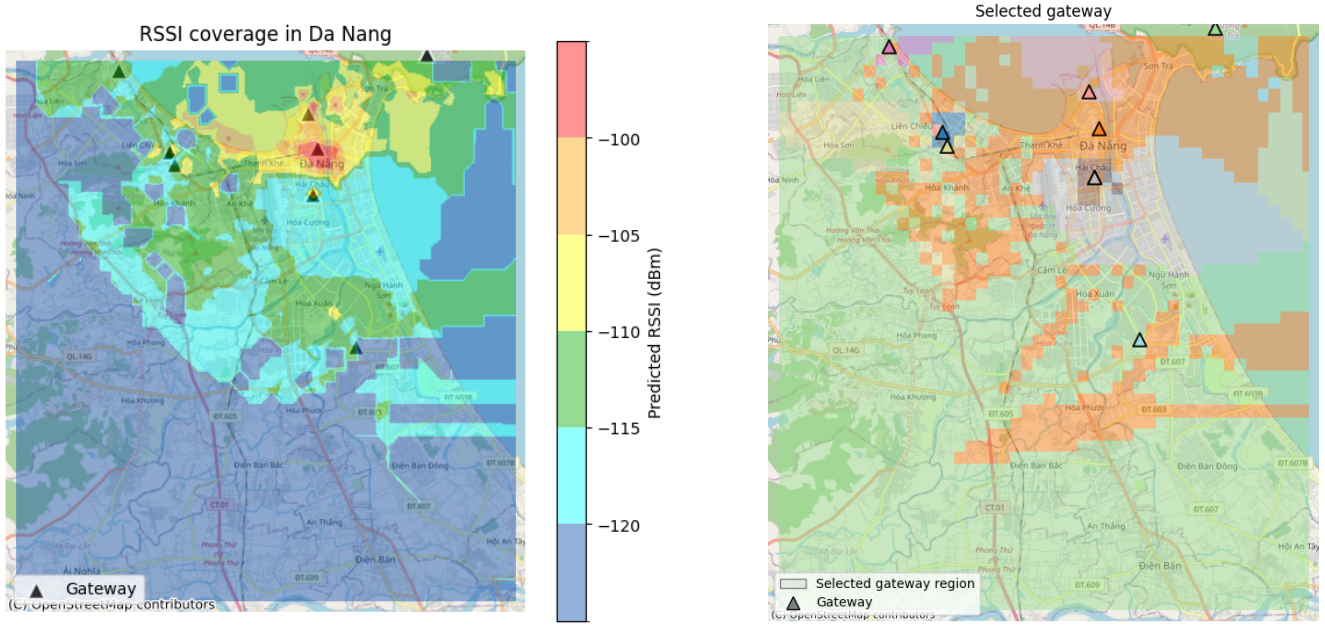


Fig. 5. (Left) Predicted RSSI coverage map over Da Nang generated on a 50 m grid. (Right) Gateway assignment per prediction location based on nearest-reference-measurement selection.

TABLE X
PREDICTION PERFORMANCE ON THE INDEPENDENT MEASUREMENT CAMPAIGN

Metric	Value
Number of measurements	~800
Measurements using unseen gateway	~600
MAE (dBm)	3.31
RMSE (dBm)	4.60

representation remains reasonably robust to moderate gateway location discrepancies.

G. Coverage Map Generation

Figure 5 illustrates an example of coverage map generation over the Da Nang study area using the proposed inference engine. RSSI values are predicted on a regular 50 m spatial grid, enabling continuous estimation of signal strength across the region rather than restricting predictions to measurement locations.

The left panel shows the resulting RSSI coverage map, where spatial variations in signal strength are captured across urban areas and surrounding regions. The right panel depicts the gateway assignment used by the inference engine for each prediction location. When no gateway is explicitly provided, the system selects the gateway associated with the nearest reference measurement, rather than simply choosing the geographically closest gateway.

This design choice results in spatially varying gateway regions, where adjacent areas may be associated with different gateways depending on the underlying measurement distribution. These transitions reflect the actual network attachment behavior observed in the dataset rather than purely geometric proximity.

The reliability of predictions depends on the spatial density of the available measurements. Regions with sparse coverage may exhibit higher uncertainty in the estimated RSSI values.

H. Practical Deployment Considerations

Unlike many machine learning studies that focus exclusively on prediction accuracy, this work also addresses deployment constraints.

The final prediction engine is completely separated from the training framework. During initialization, only the artifacts produced during offline training are loaded into memory, including the trained model, gateway database, reference measurements, and terrain datasets.

Feature generation is performed on demand for arbitrary GPS coordinates, after which RSSI predictions are produced directly by the trained model. No retraining, dataset generation, or preprocessing operations are required.

This architecture enables straightforward integration into interactive wireless coverage mapping applications. Since the trained model is loaded only once, repeated predictions can be generated without retraining or rebuilding the spatial indexes. The main computational cost is associated with feature generation, particularly the extraction of terrain-related descriptors.

I. Limitations

Although the proposed framework achieves high prediction accuracy, several limitations should be acknowledged.

First, the measurement campaigns were conducted along predefined routes rather than uniformly covering the study areas. Consequently, measurements are spatially clustered, with many samples collected only a few dozen meters apart. Although nearest-neighbor descriptors are generated without

information leakage between training and validation folds, this spatial sampling pattern may still favor the learning of highly localized propagation characteristics. As a result, the reported performance should be interpreted in the context of the geographical distribution of the available measurements.

Second, the proposed approach relies on external geographic information, including Digital Elevation Models (DEM) and OpenStreetMap land-use data, to construct terrain-related descriptors. Applying the framework to a new geographical region therefore requires the availability of equivalent GIS datasets so that the same feature extraction process can be reproduced.

Finally, most of the computational cost during deployment is associated with feature generation rather than machine learning inference itself. In particular, obstruction and Fresnel-zone descriptors require sampling the propagation path every 30 m between the gateway and the query location, which dominates the inference time when generating dense coverage maps. While the trained Extra Trees model predicts RSSI values in a negligible amount of time, feature extraction represents the dominant computational cost when generating large-scale coverage maps.

VI. CONCLUSION

This paper presented a machine-learning-based RSSI prediction framework for LoRaWAN coverage estimation.

The proposed approach combines radio configuration parameters, geometric descriptors, terrain characteristics, Fresnel-zone indicators, land-use information, and nearest-neighbor measurement statistics within a unified Extra Trees regression model.

Experiments conducted on a dataset containing more than 10,000 measurements collected from 13 gateways across two Vietnamese cities demonstrated strong predictive performance, achieving an average cross-validation score of $R^2 = 0.9285$, a MAE of 1.93 dBm, and a RMSE of 2.92 dBm.

Feature importance analysis revealed that environmental context and nearest-neighbor information play a major role in RSSI prediction accuracy, while terrain and Fresnel-related features provide complementary improvements.

In addition, an independent measurement campaign involving approximately 800 unseen measurements, including a large proportion acquired through a previously unseen gateway, demonstrated that the proposed framework remains effective under realistic deployment conditions, achieving a MAE of 3.31 dBm despite the gateway mismatch.

The final prediction engine was separated from the training framework and implemented as an inference module capable of generating RSSI estimates from only a geographical position, radio parameters, and gateway information. This architecture enables efficient integration into interactive wireless coverage mapping systems.

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